Experiment -2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1– HOME / START SCREEN

MEMORY TIDES

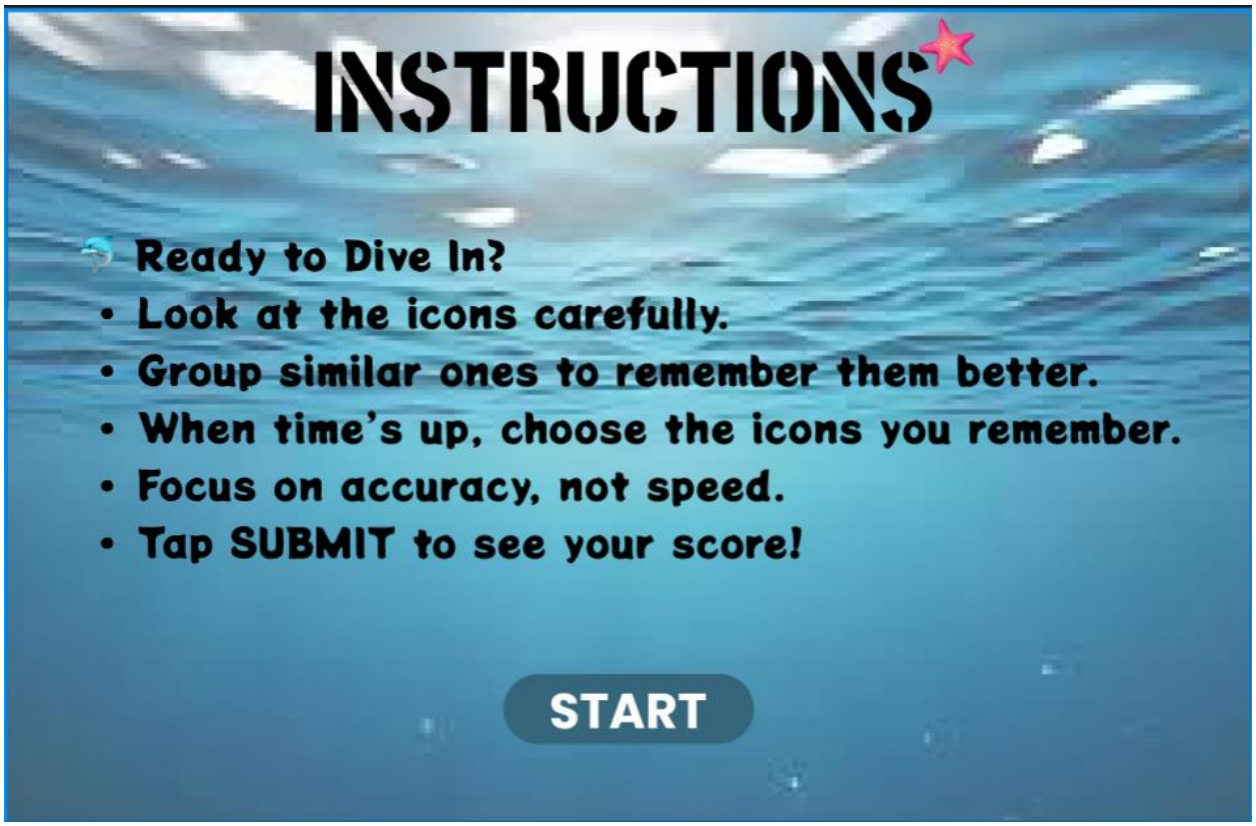
Dive into your memory.with ocean vibes.The clear view of dolphins remind the memory game very well since they one of the intelligent sea creature .it remembers rhythms and paths.

Button:
START

FRAME 2:

INSTRUCTIONS PAGE

1. **Clear and Simple Instructions**
   - The instructions are shown as short bullet points.
   - Each point explains only one action, so users can understand quickly.
2. **Step-by-Step Flow**
   - Users are first told to observe the icons.
   - Then they are guided to remember them.
   - Finally, they are informed about selecting the remembered icons.
   - This sequence avoids confusion and makes the task flow clear.
3. **Logical Information Grouping**
   - Observation → Memorization → Recall
   - These stages are naturally separated through spacing and content order.
   - This helps users mentally prepare before starting the game.
4. **Clear Visual Structure**
   - A bold **INSTRUCTIONS** heading tells users what the screen is about.
   - Proper spacing between lines prevents overcrowding.
   - The **START** button is highlighted and placed clearly for easy navigation.
5. **Time Awareness**
   - Users are informed that the task is time-based.
   - This sets expectations and encourages focused attention.



FRAME 3:

CHUNKING PHASE

Chunking Analysis of the Chunking Phase Screen

The Chunking Phase is the core stage of the **Memory Tides** game, where users visually observe and mentally organize icons for later recall. This screen applies the chunking principle through layout, visuals, and time constraints.

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1. Purpose of the Screen

- This screen represents the **visual encoding phase** of the memory task.
- Users observe a set of ocean-themed icons within a limited time.
- The goal is to encourage users to group similar icons mentally for better recall.

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2. Grouping of Visual Elements

- Icons are arranged in a grid format, making visual scanning easier.
- Similar icons (such as shells, water drops, sea plants, and animals) appear repeatedly.
- Repetition of related icons supports category-based chunking.

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3. Use of Color and Shape for Chunking

- Each icon is placed inside a rounded tile, maintaining visual consistency.
- Different background colors help users visually distinguish and group icon types.
- This color-based grouping reduces cognitive effort during memorization.

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4. Time Constraint Reinforcement

- A visible countdown timer (05 seconds) is displayed on the screen.
- The timer creates urgency and limits observation time, encouraging focused attention.
- This controlled time window strengthens short-term memory encoding.

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5. Visual Hierarchy and Focus

- The game title **"MEMORY TIDES"** is positioned at the top, maintaining context.
- The icon grid is centrally placed to draw primary attention.
- Other UI elements are kept minimal to avoid distraction during memorization.

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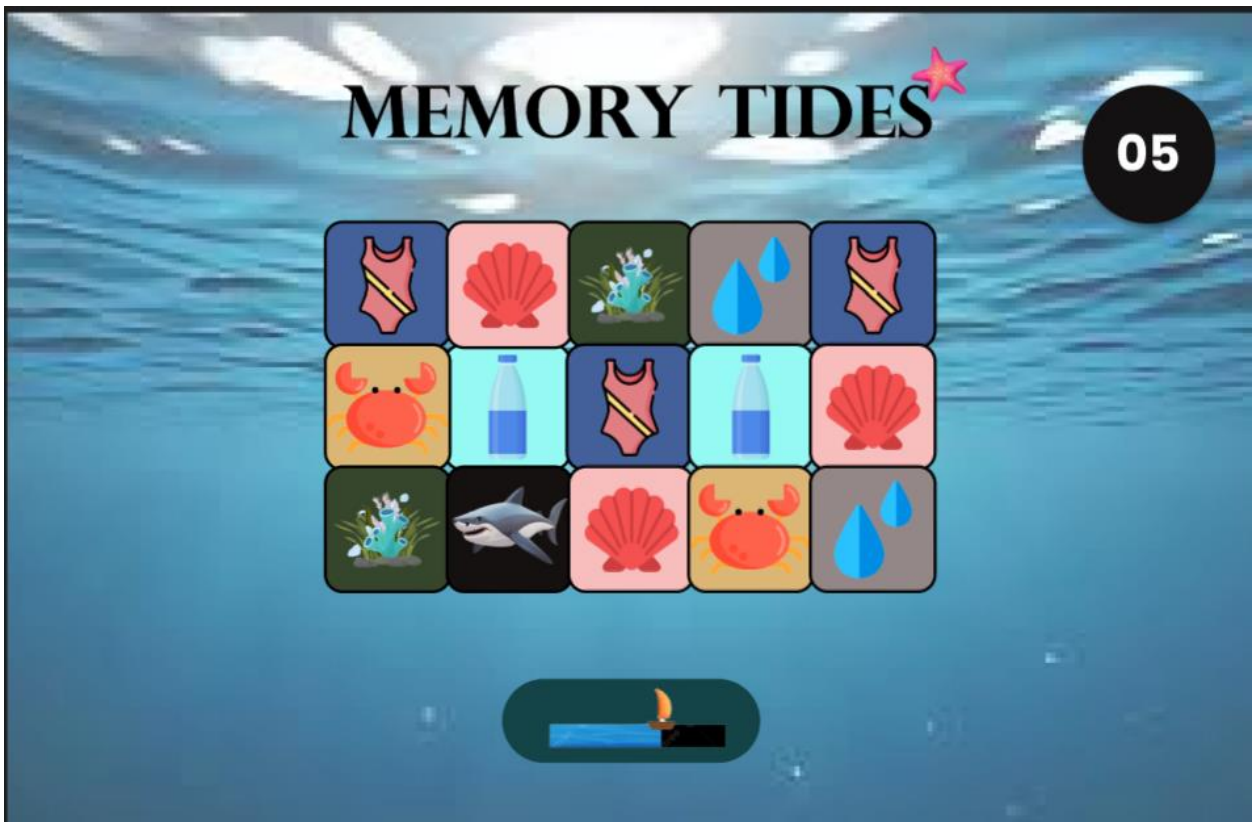
6. Cognitive Benefit

- Chunking allows users to remember groups of icons instead of individual items.
- This reduces cognitive load and improves recall accuracy in the next phase.



FRAME 4:

RECALL PHASE

Analysis of the Memory Recall Task – Selection Phase

This screen represents the **Recall Phase** of the **Memory Tides** game, where users recall and select the icons they remember from the previous Chunking Phase. Below is a breakdown of its components:

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1. Purpose of the Screen

- This screen represents the **memory retrieval stage**, where users recall information stored during the chunking phase.
- The goal is to evaluate the effectiveness of chunking and short-term memory retention.

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2. Key Elements and UI Components

Title & Instructions
- **"MEMORY TIDES"** (Bold Header) – Reinforces the identity and context of the task.
- **"Select what you remember:"** – Clearly instructs users to choose icons they recall from the previous screen.

Grid of Icon Choices
- A set of icons is displayed as selectable options.
- Some icons were shown in the Chunking Phase, while others are **distractor icons** introduced to test recall accuracy.
- Users must identify and select only the icons they remember seeing earlier.
- Each icon includes a checkbox below it to allow user selection.

🐬 Dolphin Theme Elements
- Ocean-themed visuals and dolphin graphics maintain visual consistency with earlier screens.
- These elements add a playful and engaging feel without distracting from the recall task.

"SUBMIT" Button (Bottom Center)
- Once users complete their selections, they press **SUBMIT** to confirm their answers.

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3. How the Recall Phase Works

1. Users observe the icons displayed on the screen.
2. They mentally recall which icons appeared in the Chunking Phase.
3. Users select the remembered icons using the checkboxes.
4. Distractor icons test whether users can accurately distinguish remembered items from new ones.
5. Clicking **SUBMIT** finalizes the selection and moves the user to the result screen.

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4. Cognitive and UX Benefits

- Tests the accuracy of short-term memory recall.
- Distractor icons help measure the effectiveness of chunking.
- Clear selection mechanisms (checkboxes) make interaction simple and user-friendly.
- The ocean-themed design keeps users engaged while maintaining focus on the task.

FRAME 5:

RESULT PAGE:

Analysis of the Memory Recall Task – Score & Feedback Screen

This screen represents the **Score & Feedback Phase** of the **Memory Tides** game, where users receive their performance evaluation based on the selections made in the previous Recall Phase.

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1. Purpose of the Screen

- Provides feedback on recall accuracy by displaying the number of correct answers.
- Allows users to decide their next action (continue, restart, or exit).

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2. Key Elements & UI Components

Title & Score Display
• **"MEMORY TIDES"** – Reinforces the game's title and context.
• **"SCORE"** – Indicates that the user's performance is being displayed.
• A score board displays the user's score (for example, **3/7**), showing correct answers out of total attempts.

Action Buttons (Right Side)

1. **"HOME"** – Returns the user to the home screen.
2. **"RESTART"** – Allows the user to retake the memory recall task.
3. **"EXIT"** – Ends the task and exits the game.

🐬 **Dolphin-Themed Visuals**
• Dolphin and ocean-themed illustrations add a playful, gamified element.
• These visuals maintain consistency with earlier screens and enhance user engagement.

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3. How This Phase Works

1. The system evaluates the user's selections from the Recall Phase.
2. It calculates the recall score and displays it on the score board.
3. Users review their performance and choose their next action:
    - **Home** if they want to return to the main screen.
    - **Restart** if they want to improve their recall.
    - **Exit** if they want to stop playing.

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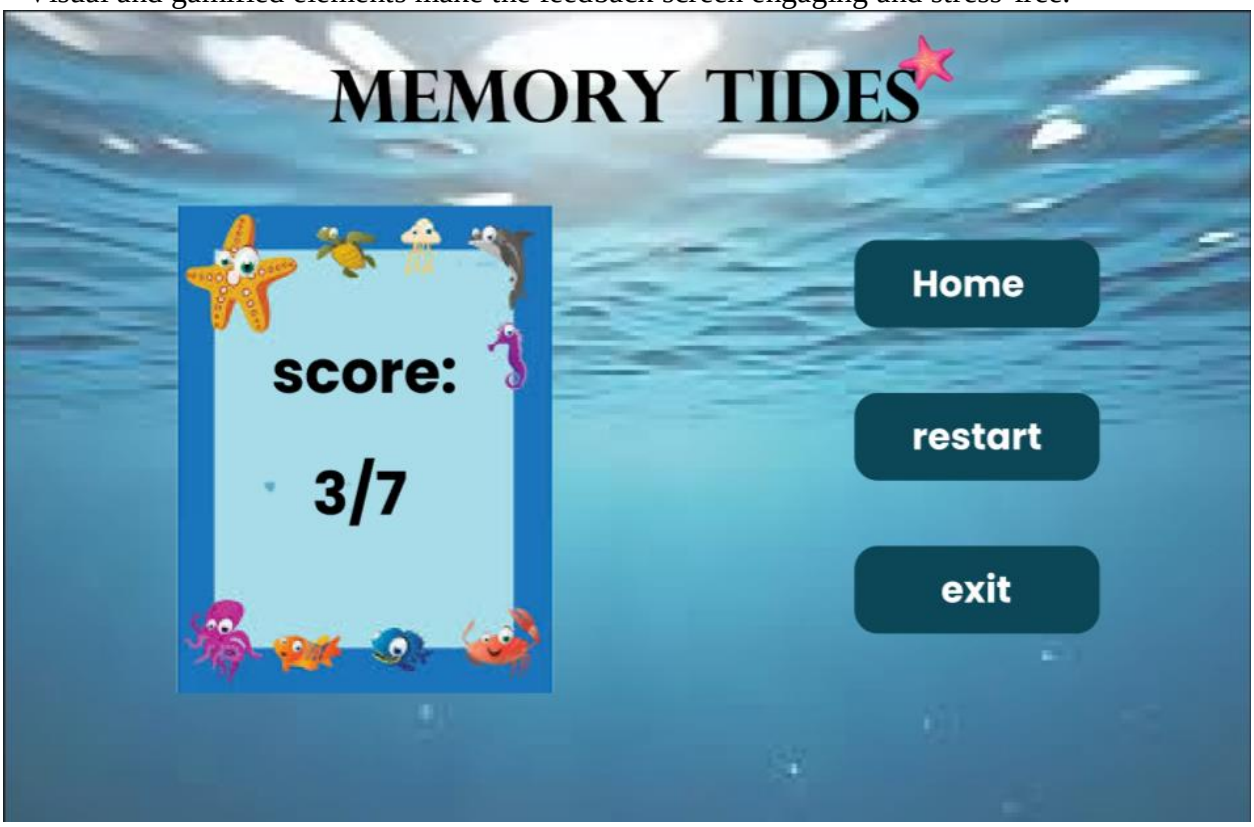
4. Cognitive & UX Benefits

• Instant feedback helps users understand their memory performance.
• Multiple options (Home, Restart, Exit) give users control over their experience.

• Visual and gamified elements make the feedback screen engaging and stress-free.



PROTOTYPE LINK:

https://www.figma.com/proto/itbRAXag3FHiTbhdaxBtNJ/Untitled?node-id=0-1&t=HA1V5hP35lcRn5jz-1